Evaluating Prompting Strategies and Token Efficiency in Generative AI Models

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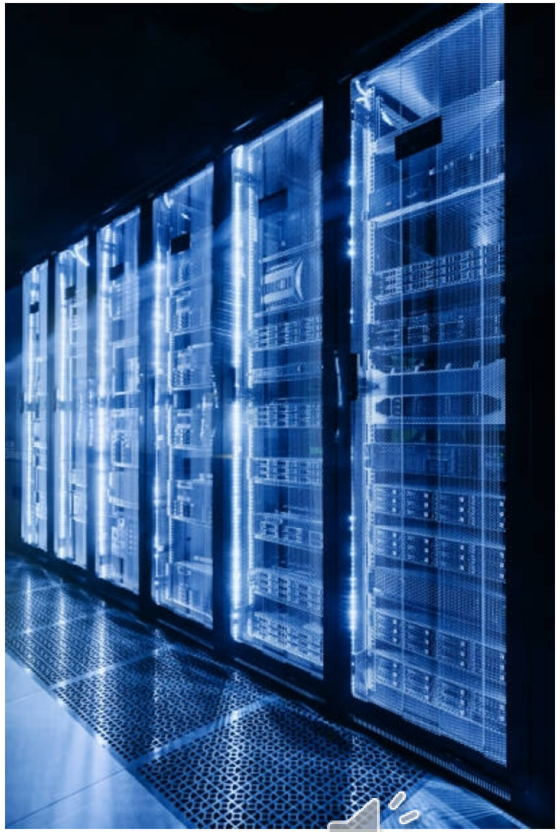
Background – The Hidden Environmental Cost of AI

Sources: Beck 2023; Morrison 2024; KCRG-TV9 2024; Kaufman 2023; Li et al. 2023.

- **High Energy Consumption:** Energy of model inference now outweighs training for many widely used systems
- **GPU Production:** Growing demand for AI hardware and data centers
- **Misalignment:** General lack of accountability in prompting practises and environmental impact
- **Lack of Transparency:** Users rarely see the environmental cost embedded in each prompt



Case study: Microsoft AI data centers in West Des Moines

- Microsoft's West Des Moines data center cluster, used to train GPT-4, consumed **11.5 million gallons of water in a single month** during the model's peak training period. By 2024, Microsoft's five campuses used **68.5 million gallons**, making the company **the city's largest water user**.

- At the same time, central Iowa faced **record-high nitrate levels** in the Raccoon and Des Moines Rivers, overwhelming treatment capacity and prompting the region's **first-ever lawn-watering ban** and increased reliance on bottled water.

- Although the ban was triggered primarily by nitrate contamination, the rapid expansion of AI-driven data centers compounds local stress: Microsoft is now the **largest single customer** of West Des Moines Water Works and is planning continued expansion, locking in high long-term withdrawals unless cooling technology shifts to lower-water systems.

- This case highlights how **LLM-driven compute demand amplifies environmental pressure**, especially in regions already facing water-quality and infrastructure challenges. (Morrison 2024)

Research Questions and Experiment Design

RQ1: Do more elaborate prompts increase accuracy on math word problems?

RQ2: How do different models compare in accuracy under identical prompt conditions?

RQ3: Which prompting strategy is most token-efficient?

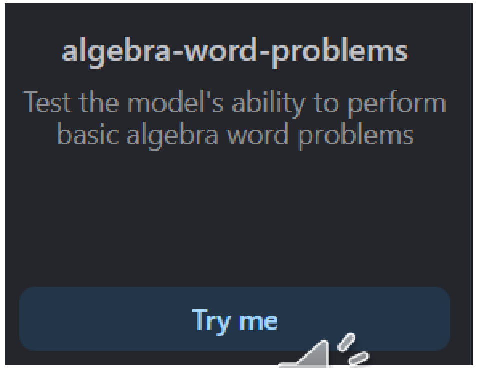
Variation in prompts: 10 repetitions each

Chain-of-Thought Prompt
Prompt: “Let’s solve this step by step… show each step, then give the final numeric answer.”
Type: Structured reasoning (Chain-of-Thought) + output formatting.

Expert Persona Prompt
Prompt: “You are an experienced high-school math teacher… explain step by step, then give the final answer.”
Type: Expertise framing (persona) + structured reasoning + output formatting.

Constrained Output Prompt
Prompt: “Solve the following math word problem. Give only the final numeric answer, with no explanation.”
Type: Output constraint / suppression of reasoning steps. (Shanahan et al. 2023)

ChainForge



Experiment Design – Token Efficiency by Model

Comparing models for token efficiency and accuracy

GPT-5-mini

Represents a frontier-leaning baseline.

Demonstrates strong reasoning and instruction-following performance. (OpenAI 2024)

GPT-5-nano

Significantly size-constrained version of the 5-series architecture

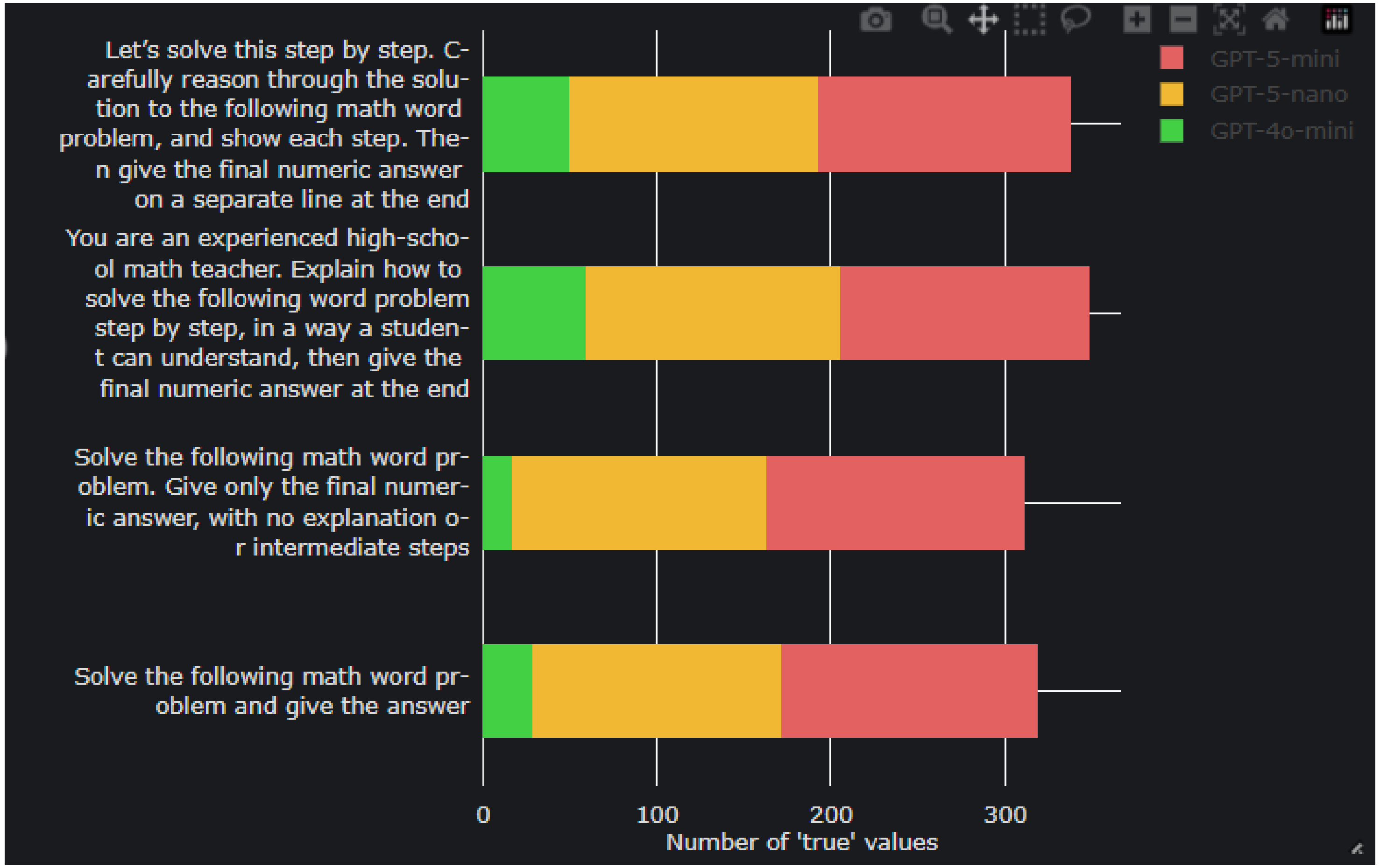
Used to examine how prompting strategies behave under minimal compute generation. (OpenAI 2024)

GPT-4o-mini

Represents the previous "omni" generation of models and provides a mid-capability baseline

Allows comparison across generations (4o vs 5-series) capacity or model architecture. (OpenAI 2024)

Results



GPT 5-mini consistently achieved the highest number of correct answers.

Followed by GPT5 nano and GPT 4 nano being the weakest.

Models performed best when they were allowed to reason step-by-step: chain-of-thought and teacher-persona prompts produced the highest accuracy across all architectures.

Smaller models suffered the most under “answer-only” prompting, confirming that reasoning scaffolds matter even more when computational capacity is limited.

Results

Model	Prompt Strategy	Avg Prompt Tokens	Avg Total Tokens
GPT-4o-mini	Let's solve this step by step… (COT)	186.98	187.98
GPT-4o-mini	Solve the following math word problem and give… (medium explicit)	154.98	155.98
GPT-4o-mini	Solve the following math word problem. Give only… (minimal instruction)	166.98	167.98
GPT-4o-mini	You are an experienced high-school math teacher… (role+reasoning)	188.31	189.31
GPT-5-mini	Let's solve this step by step…	186.98	187.98
GPT-5-mini	Solve the following math word problem and give…	154.98	155.98
GPT-5-mini	Solve the following math word problem. Give only…	166.98	167.98
GPT-5-mini	You are an experienced high-school math teacher…	188.31	189.31
GPT-5-nano	Let's solve this step by step…	186.98	187.98
GPT-5-nano	Solve the following math word problem and give…	154.98	155.98
GPT-5-nano	Solve the following math word problem. Give only…	166.98	167.98
GPT-5-nano	You are an experienced high-school math teacher…	188.31	189.31

Sustainability Ranking:

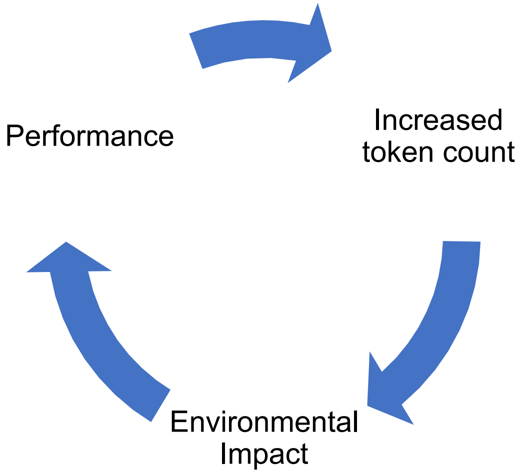
GPT-5-mini (0.00514) – Most sustainable

GPT-5-nano (0.00457) – Moderately sustainable

GPT-4o-mini (0.00143) – Very low sustainability

Reflection and Conclusions

- Prompt structure clearly shaped model behaviour: chain-of-thought and expert persona prompts produced longer, more detailed reasoning, while constrained-output prompts sharply reduced verbosity and inference steps
- Token usage and energy implications varied by prompt type: prompts demanding explanations generated significantly more tokens, confirming that user-facing prompt design directly affects the environmental footprint of LLM use
- Consistency vs accuracy trade-offs emerged: step by step and teacher-persona prompts tended to improve reasoning clarity but “answer only” prompts were faster, lighter, and more efficient. This highlights a tension between performance and sustainability



Small changes in how we prompt LLMs create big differences in reasoning style, token usage, and environmental impact.

Thank You For Listening!

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